

AI Radio Cleaner — Feature Spec

Part of: Stemy Pro Plan (\$24.99/month) **Status:** New feature to build **Position:** Third core feature alongside Stem Mixing Console and Reference Matching

What It Does (Simple Version)

Customers upload a song with explicit/curse words. The AI automatically detects every curse word in the vocal. The customer chooses HOW to censor each curse word (one of four effects). The AI applies the chosen effect. The customer downloads a clean "radio edit" version of their song.

Why customers want this:

- Get songs played on radio (FM/AM stations require clean edits)
 - Get songs accepted for TV / movies / commercials
 - Get into curated Spotify / Apple Music playlists (clean versions only)
 - Avoid getting flagged on TikTok / Reels
 - Pro mastering studios charge \$50-\$150 per radio edit. Stemy does it automatically.
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How It Works (User Flow)

Step 1 — Upload Song

- Customer uploads finished song (WAV, MP3, FLAC, M4A)
- Same upload limits as rest of Stemy (100 MB)

Step 2 — AI Detects Curse Words

- AI scans the vocal track
- Uses speech-to-text (Whisper API or AssemblyAI) to transcribe lyrics
- Finds curse words from a profanity database

- Returns a list of detected curse words with timestamps:

```
[  
  { word: "fuck",   startTime: 12.34, endTime: 12.62 },  
  { word: "shit",  startTime: 28.10, endTime: 28.41 },  
  { word: "bitch", startTime: 45.78, endTime: 46.15 },  
  ...  
]
```

Step 3 — Customer Picks Censor Effect

Customer sees a list of detected curse words. For EACH word, customer picks one of 4 effects (or applies one effect to all):

Effect 1: MUTE

- The word is silenced (volume drops to zero during the word)
- Cleanest option, sounds like the word was never there
- Best for: tight clean edits

Effect 2: REVERSE

- The audio of just the word is played backwards
- Creative effect, popular in hip-hop
- Best for: keeping rhythm/flow alive

Effect 3: TAPE STOP

- Pitch and speed slow down rapidly during the word (like an old tape deck stopping)
- Dramatic, modern effect
- Best for: songs with edge

Effect 4: TURNTABLE BACKSPIN

- Sounds like a DJ scratching/spinning the record back
- Rhythmic and stylish
- Best for: hip-hop and dance tracks

Step 4 — Process & Download

- Customer clicks "Generate Clean Version"

- AI applies the chosen effect to each detected word
- Returns final clean WAV + MP3
- Customer downloads radio-ready clean edit

Step 5 — Optional: Manual Override

- Customer can preview each detected word before processing
 - Customer can mark false positives ("don't censor this word")
 - Customer can add words manually if AI missed any
 - Customer can change the effect per word individually
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Backend Requirements

Speech-to-Text

- **OpenAI Whisper API** (recommended — most accurate, \$0.006/minute)
- OR AssemblyAI (also good)
- OR self-host Whisper (free but needs GPU)

Profanity Detection

- Use a profanity word list database (multiple available open-source)
- Match transcribed words against the list
- Allow customer to add custom words to censor (e.g., names, slurs, brand names)
- Default profanity list should include: standard FCC-banned words plus common variations

Audio Processing (the 4 effects)

Use **FFmpeg** or **Python (pedalboard, librosa, sox)** to apply effects.

MUTE effect:

```
ffmpeg -i input.wav -af "volume=enable='between(t,12.34,12.62)':volume=0" output.
```

Apply gentle 5ms fade-in/out to avoid clicks.

REVERSE effect:

- Cut the segment between startTime and endTime
- Reverse the audio buffer
- Splice back in
- Apply 5ms crossfades on both sides

TAPE STOP effect:

- Apply ramping pitch + speed reduction over the duration of the word
- Pitch drops from 0 semitones to -12 semitones
- Speed drops from 1.0x to 0.0x
- Add slight low-pass filter sweep for tape feel
- Use sox or pedalboard's PitchShift + a custom envelope

TURNTABLE BACKSPIN effect:

- Reverse playback at increasing speed
- Add vinyl scratch noise (pre-loaded sample)
- Quick speed ramp: starts at -2x, accelerates to -4x backward
- Mix in scratch noise sample

File Output

- Output format: 24-bit WAV (primary) + 320kbps MP3 (secondary)
- Apply same -14 LUFS / -1 dBTP standard as rest of Stemy
- Embed metadata noting "[CLEAN VERSION]" tag

Database Schema Addition

```
clean_jobs (  
  id uuid primary key,  
  user_id uuid references users(id),  
  master_id uuid references masters(id),  -- nullable, can run on standalone trac  
  source_url text,  
  output_url text,
```

```
detected_words jsonb, -- array of {word, startTime, endTime, effect}
status text, -- queued, transcribing, censoring, complete, failed
total_words_detected int,
total_words_censored int,
default_effect text, -- "mute" | "reverse" | "tapestop" | "backspin"
created_at timestamp,
completed_at timestamp
)
```

API Endpoints

POST	/api/clean/upload	(customer uploads audio file)
POST	/api/clean/:id/transcribe	(run AI detection)
GET	/api/clean/:id/words	(get list of detected words)
PATCH	/api/clean/:id/words/:wordId	(change effect for one word)
POST	/api/clean/:id/process	(generate final clean version)
GET	/api/clean/:id/download	(signed URL to download)
GET	/api/clean	(list user's clean jobs)

UI Layout (already designed in mockup file)

The standalone AI Radio Cleaner page has 4 stages:

Stage 1: Upload zone (similar to Quick Master upload)

Stage 2: Analyzing (loading animation while AI transcribes)

Stage 3: Word list editor

- Header: "AI detected X curse words"
- Default effect selector (apply to all)
- Scrollable list, each row shows:
 - Timestamp (0:12.34)
 - Detected word (highlighted red)
 - Surrounding lyrics for context
 - Effect dropdown (Mute / Reverse / Tape Stop / Backspin)

- Preview button (▶ icon)
- Skip checkbox ("don't censor this one")
- "Add Manual Word" button at bottom
- Big "Generate Clean Version" button

Stage 4: Result

- Success screen with playback comparison (Original vs Clean)
 - Download buttons (WAV + MP3)
 - "Detected and censored 12 words" stat
 - "Master Another Track" button
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Pricing Logic

- AI Radio Cleaner is **Pro Plan only** (\$24.99/month)
 - Unlimited radio edits per month (matches the rest of Pro)
 - Basic users see a teaser with "Upgrade to Pro" wall
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Marketing Angle

Three reasons this feature sells the Pro plan:

1. **Save money:** Pro studios charge \$50-150 per radio edit. Stemy unlimited for \$24.99.
2. **Save time:** Manual radio edits take 1-2 hours. Stemy does it in 90 seconds.
3. **Open new platforms:** Get on radio, TV, ads, clean playlists — without re-recording.

Tagline ideas:

- "From explicit to radio-ready in 90 seconds"
 - "Get on the radio, keep your bars"
 - "Clean edits that actually sound clean"
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Build Order (suggested)

Phase 1 (MVP):

1. Upload + transcription via Whisper API
2. Profanity word list matching
3. Just MUTE effect working
4. Download output
5. UI for word list with mute toggles

Phase 2: 6. Add REVERSE effect 7. Add TAPE STOP effect 8. Add TURNTABLE BACKSPIN effect 9. Per-word effect picker 10. Preview button for each word

Phase 3: 11. Manual word addition 12. False positive correction 13. Custom profanity lists per user 14. Batch processing multiple songs

Estimated Cost Per Customer Use

- Whisper transcription: ~\$0.006/minute → ~\$0.02 per 3-min song
- Audio processing: server compute, negligible
- Storage: ~\$0.001 per file
- **Total cost to Stemy per radio edit: ~\$0.03**
- Customer pays \$24.99/month for unlimited
- Even if heavy user does 100 radio edits per month = \$3 cost vs \$24.99 revenue

Profitable feature.

Frontend UI mockup is in separate HTML file. See standalone interactive demo for visual design and animations.